

Juan Manuel

Espejo Salcedo

Curriculum Vitae

Munich, Germany
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My research focuses on the observational study of galaxy formation and evolution, particularly through infrared observations of galaxy morphology and kinematics at cosmic noon.

GENERAL INFORMATION

Name Juan Manuel Espejo Salcedo.
Date of Birth 13th March, 1993.
Nationality Colombian.
Occupation Postdoctoral researcher, Max Planck Institute for Extraterrestrial Physics.
Website <https://juancho9303.github.io>
YouTube <https://www.youtube.com/c/betweenwhiles>

EDUCATION - WORK

2023- **Postdoctoral researcher**, *Galaxy Evolution*, Infra-Red Group - Max Planck Institute for Extraterrestrial Physics (MPE), Garching, Bayern bei München, Germany
2019-2023 **PhD in Astronomy**, *Thesis: The Role of Angular Momentum in the Evolution of Star-forming galaxies*, Swinburne University of Technology, Melbourne, Australia
2016-2018 **M.Sc. in Astronomy and Cosmology**, *Sterrenwacht*, Leiden University, Netherlands
Thesis 2018 “Phenomenology of Large Scale Structure in scalar-tensor theories: joint prior covariance of w_{DE} , Σ and μ in Horndeski”. Prof. Alessandra Silvestri
Thesis 2017 “Study of BCG subtracted images of nearby clusters”. Prof. Henk Hoekstra
2011-2016 **B.Sc. in Astronomy**, *Physics Institute, Universidad de Antioquia*, Medellín, Colombia.
Thesis “Mass Modelling of Globular Clusters in the Milky Way”. Prof. Juan Carlos Muñoz Cuartas
2010 **High School**, *Colegio Seminario Diocesano*, Duitama, Colombia.

ADDITIONAL EDUCATION

2019-2021 **Graduate Certificate of Research and Innovation Management**, *Swinburne University of Technology, Melbourne, AU.*
2011-2015 **Extension Courses: Black Holes**, Prof. Jorge Zuluaga, **Radiative Processes in Astrophysics**, Prof. Elena Rossi, **Observational Techniques in Astronomy**, Prof. Bernard Brandl, **Interferometry**, Prof. Rudolf Lepole, **Gravitational Lensing**, Prof. Konrad Kuijken, **Galaxy Formation**, Jarle Brinchmann, **Introduction to Radio Astronomy**, Prof. Stanley Kurtz.
2018-2009 **11th grade in American High School as a foreign exchange student**, *Rowan County Senior High School*, Morehead, Kentucky, United States.

FIELDS OF INTEREST

Galaxies and Cosmology. Large-scale structure formation. Galactic astrophysics. Computational astrophysics. General astrophysics and physics.

LANGUAGES

Spanish Native speaker.
English Fluent.
Portuguese Basic.

COMPUTER SKILLS

Systems Linux, MacOS, Windows.
Development Python, C, Bash, SQL, ssh. I developed a Python-based GUI for ERIS data reductions.
Software DysmalPy, git, Galfit, Starlight, Iraf, Mathematica, LaTeX, gnuplot, SExtractor, Adobe Premiere Pro, EFTCam.
Tools Experience with supercomputers (NOVAMARIS at Leiden, OzStar at Swinburne), MonteCarlo, Numerical methods, emcee.
Repositories A list of some of my projects can be found in my *github* page: <https://github.com/juancho9303>.

HONOURS, AWARDS, AND ACCOMPLISHMENTS

2022 Swinburne High-degree Research Boost Award, 2000 AUD one-off payment, *Swinburne University of Technology, Melbourne, Australia*.
2018-2022 Swinburne University Postgraduate Research Award (SUPRA), *Swinburne University of Technology, Melbourne, Australia*.
2016-2018 Leiden Excellence Scholarship (LExS), *Leiden University, Leiden, Netherlands*.
2016-2018 Colfuturo postgraduate funding, Bogotá, Colombia.
2016 Enlazamundos funding for development grant, *Sapienza, Medellín, Colombia*.
2014 Top-ranked student (4th year, Astronomy program), tuition fully waived. *Universidad de Antioquia, Medellín, Colombia*.
2012-2013 Basketball merit scholarships, full tuition waiver. *Universidad de Antioquia, Medellín, Colombia*.
2005-2009 Valedictorian (1st in class), full tuition waiver. *Colegio Seminario Diocesano, Duitama, Colombia*.

PUBLICATIONS (first-author only)

2026 MSA-3D: Rotation Curves and Dark Matter Fractions at $z \sim 0.5-1.7$ with JWST/NIRSpec. *Juan M. Espejo Salcedo, Danail Obreschkow, Karl Glazebrook, Tucker Jones, Ivana Barišić, Natascha M. Förster Schreiber, Takafumi Tsukui, Xin Wang, Mengting Ju, Qianqiao Zhou, Amit Nestor-Schachar, Ryan L. Sanders, Stavros Pastras, Namrata Roy, Alaina Henry, Kyle Westfall, Themiyá Nanayakkara, Matthew Malkan, Fahmi M. Al Farisy, Isaac Kanowski*. arXiv:2606.27853

- 2025 Galaxy Morphologies at Cosmic Noon with JWST: A Foundation for Exploring Gas Transport with Bars and Spiral Arms.** *Juan M. Espejo Salcedo, Stavros Pastras, Josef Vácha, Claudia Pulsoni, Reinhard Genzel, Natascha M. Förster Schreiber, Jean-Baptiste Jolly, Capucine Barfety, Jianhang Chen, Giulia Tozzi, Daizhong Liu, Lilian Lee, Stijn Wuyts, Linda Tacconi, Ric Davies, Hannah Übler, Dieter Lutz, Emily Wisnioski, Jinjy Shangguan, Minju Lee, Sedona Price, Frank Eisenhauer, Alvio Renzini, Amit Nestor Shachar, Rodrigo Herrera-Camus.* 2025A&A...700A..42E
- 2025 A Shallow Slope for the Stellar Mass–Angular Momentum Relation of Star-forming Galaxies at $1.5 < z < 2.5$.** *Juan M. Espejo Salcedo, Karl Glazebrook, Deanne B. Fisher, Sarah M. Sweet, Danail Obreschkow, Natascha M. Förster Schreiber.* 2025MNRAS.536.1188E
- 2022 Multi-resolution Angular Momentum Measurements of $z \sim 1.5$ –2 Star-forming Galaxies.** *Juan M. Espejo Salcedo, Karl Glazebrook, Deanne B. Fisher, Sarah M. Sweet, Danail Obreschkow, A. M. Swinbank, Steven Gillman, Alfred L. Tiley.* 2022MNRAS.509.2318E
- 2019 Phenomenology of Large-scale Structure in Scalar–Tensor Theories: Joint Prior Covariance of w_{DE} , Σ , and μ in Horndeski.** *Juan M. Espejo Salcedo, Simone Peirone, Marco Raveri, Levon Pogosian, Alessandra Silvestri.* Phys. Rev. D 99, 023512

WORK EXPERIENCE

- 2023-2026** Three observing runs at the VLT, with adaptive optics assisted observations with ERIS at UT4.
- 2023-2025** Tea meeting coordinator - Infra-red group, MPE.
- 2024** Summer internship supervisor - Josev Vácha, Oxford University.
- 2022** AstroTour coordinator at Swinburne University of Technology.
- 2020-2021** Awarded observing time as Co-I for Keck (OSIRIS) twice, ALMA (cycle 8), and JWST (cycle 1).
- 2021** Tutor for the “Discovering the Universe” undergrad unit at Swinburne University of Technology.
- 2020-2021** Mentor of colombian students looking for post-graduate positions as part of RECA mentoring program.
- 2020-2021** Student representative of the Swinburne Time Allocation Committee for the Keck Telescope (STACK).
- 2020** Tutor for the “Introduction to physics” undergrad unit at Swinburne University of Technology.
- 2019-2022** Public AstroTours at Swinburne University of Technology.
- 2015** English Tutor online **NiceTalk Tutor app, Enter Trainer app.**
- 2014-2015** Science development magazine editor **"Mínima Acción" Magazine**, Medellín, Colombia.
- 2014** **Stratospheric Cosmic Ray Detector development**, *Group of computational physics and astrophysics (FaCom), Ideatech (Ruta N) & Universidad de Antioquia*, Medellín, Colombia.
- 2014, 2015** English and Math Tutor.

PARTICIPATION IN EVENTS

- 2026** (Contributed talk) **MSA-3D: Rotation curves and dark matter fractions at $z \sim 1$ with JWST/NIRSpec**, *How stars move around and what they tell us about galaxies' masses, formation and evolution*, Sexten, Italy, July 2026.
- 2026** (Contributed talk) **Galaxy Morphologies at Cosmic Noon with JWST**, *Beyond Hubble*, Cambridge, UK, April 2026.
- 2025** (Invited talk) **The emergence of $m=1,2,3$ modes in JWST disks**, *ALMA, JWST, Gaia – the Milky Way connection*, Paris, France, October, 2025.
- 2025** (Contributed talk) **Galaxy Morphologies at Cosmic Noon with JWST**, *12th Sino-German meeting on galaxy formation*, Chengdu, China, September 2025.

- 2025 (Contributed talk) **Non-axisymmetric structures and inflows in cosmic-noon disks**, *EAS annual meeting*, Cork, Ireland, June 2025.
- 2025 (Seminar talk) **Galaxy Morphologies at Cosmic Noon with JWST: A Foundation for Exploring Gas Transport with Bars and Spiral Arms**, *TMT Talk - ESO* Santiago, Chile.
- 2024 (Conference talk) **The angular momentum of star-forming galaxies at cosmic noon and its role in disk morphology**, *HERAS workshop (The Nature and the Dynamics of Structures Observed in Galactic Disks)* Athens, Greece.
- 2023 (Seminar talk) **The multi-resolution angular momentum measurement of star-forming galaxies at cosmic noon**, *Seminar to the IR group at MPE (Max Planck Institute for Extraterrestrial Astrophysics)* Garching, Germany, January 2023.
- 2023 (Invited talk) **Galaxies - State of the art and open questions**, *Senior Talk - XV IMPRS Symposium Program* Garching, Germany, 2023.
- 2022 (Contributed talk) **What drives disk instabilities and star-forming clumps at cosmic noon?**, *IAU GA Symposium 373: "Resolving the rise and Fall of Star Formation"*, in Busan, South Korea, August, 2022.
- 2022 (Contributed talk) **What drives disk instabilities and star-forming clumps at cosmic noon?**, *ASTRO-3D science meeting*, hosted in Melbourne, June, 2022.
- 2021 (Science Organizing Committee member) **Harley Wood School for Astronomy**, *Wodonga, VIC, AU*.
- 2021 (Contributed talk) **Multi-resolution angular momentum measurements of $z \sim 1.5 - 2$ star-forming galaxies.**, *MAVIS workshop*, Online event, hosted in Italy, July, 2021.
- 2021 (Poster presentation) **Multi-resolution angular momentum measurements of $z \sim 1.5 - 2$ star-forming galaxies.**, *Keck Science Meeting*, Online event, hosted in San Diego California, USA, 2021.
- 2020 (Contributed talk) **Improved Angular Momentum measurements of star-forming galaxies with the aid of adaptive optics**, *European Astronomical Society annual meeting EAS2020*, Leiden, The Netherlands, 2020.
- 2020 (Organizer and presenter) **JWST workshop for Australian astronomers**, *Swinburne University of Technology*, Melbourne, Australia, 2020.
- 2018 Participant **Galactic angular momentum focus meeting**, *30th General assembly of the IAU*, Vienna, Austria.
- 2014 (Oral presentation) **Stratospheric Cosmic Ray detector**, *IV Colombian Congress of Astronomy and Astrophysics*, Pasto, Colombia, 2014.

PERSONAL ACTIVITIES

Basketball. Track and Field. Writing. Teaching. Origami. Hiking. Reading. Science public outreach.

REFERENCES

(from most recent to earlier)

- **Natascha Förster Schreiber**. *Postdoc supervisor*.
Senior Staff Scientist, MPE, Garching, GE.
forster@mpe.mpg.de
- **Karl Glazebrook**. *Ph.D supervisor*.
Laureate Fellow & Distinguished Professor at the Centre for Astrophysics and Supercomputing, SUT, Melbourne, AU.

kglazebrook@swin.edu.au

- **Alessandra Silvestri.** *Master thesis (second project) supervisor.*
Lorentz Institute, Leiden University, Netherlands.
silvestri@lorentz.leidenuniv.nl
- **Juan Carlos Muñoz Cuartas.** *Undergraduate thesis supervisor*
Assistant Professor, Physics Institute, Universidad de Antioquia, Medellín, Colombia.
jcmunoz@fisica.udea.edu.co